

OpenProject2 — an ERP factory for European SMEs

A machine that turns a **tenant specification** into a running, tested, compliance-gated ERP — **spec in, ERP out** — repeatably for hundreds of small businesses. The ERP is the output; the factory is the product. **Tenant #1** is a Spanish *reformas* (renovation) company.

- ✓ P0 Foundation
- ✓ P1 Walking skeleton
- P2 Packs & UI · in progress
- P3 Pipeline
- P4 Fleet
- P5 Proof

16 green commits on the branch — every one lint + type + test + build + boundary clean	0.02s to create tenant #2 — config only, zero code. Marginal cost near zero	~70 automated tests + live-Postgres row-level-security contract tests in CI	2027 Verifactu deadline verified & gated — chaining built now, submission off until certified
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Where the build stands today

✓ DONE & CI-VERIFIED - Layered kernel · capabilities · packs architecture, boundaries enforced by the build. - Working slice: presupuesto → factura with effective-dated IVA, persisted legal justification, and tamper-evident invoice chaining. - Durable persistence on PostgreSQL with row-level security per tenant. - A first web workspace per tenant, running the same composed services.	● IN PROGRESS / NEXT - Real data-entry screens (presupuestos, partidas) & PDF export. - Spanish tax-filing data (Modelo 303/347/...), bank import, working-hours log. - Renovation certificaciones: quoted-vs-actual per line — the tenant's biggest pain. - Deploy pipeline, fleet control plane, and the timed onboarding proof.
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How to read this report. Page 2 is the architecture. Page 3 is what exists and is proven. Page 4 is the plan. **Page 5 is the one that needs you:** the decisions and inputs required from the business — each with a safe default so nothing is blocking today.

How the factory is built

Five layers, each depending only **downward**. A country (jurisdiction) and a trade (vertical) are **orthogonal** — so covering N countries × M trades costs **N + M**, never N × M. Adding Germany later is a new pack, not an edit to the core.

Plugins	Sandboxed, contract-tested escape hatch for genuinely bespoke logic	future
Tenant config	Data only — branding, price lists, series, users. Customisation is data, never a fork	✓ built
Vertical packs	Trade knowledge — <i>construction / reformas</i> : mediciones, partidas, terminology	✓ built
Jurisdiction packs	Country law — <i>es-ES</i> : IVA, IRPF, invoice chaining — effective-dated	✓ built
Capabilities	Quoting, billing... define ports . Zero country / trade knowledge	✓ built
Kernel	Tenancy, money, effective-dating, composition, persistence, events	✓ built

▲ depends downward only — enforced in CI · the kernel & capabilities contain no **IVA**, no **partida**, no country anything

PORTS & ADAPTERS — COMPOSITION AT RESOLVE TIME

Billing
capability

→

tax@1
port (contract)

→

es-ES
adapter

The tenant spec picks the packs; the resolver binds adapters to ports and **fails loudly at resolve time** on any conflict or gap — never in front of a customer.

TIME IS A DIMENSION OF CORRECTNESS

Tax rules resolve by **date**, not version: amending a March invoice in November applies **March's** law. Every decision is stored **with its justification** — legal basis, inputs, rule id — on the invoice, never recomputed.

Boundaries are enforced, not aspirational. A build-time linter fails CI if a capability imports a country pack, if one pack imports another, or if a forbidden literal (a tax name, a rate like `0.21`, a hardcoded locale) leaks into the core. It already caught — and we fixed — a real leak on its first run.

The falsifiable test (proven, runs in CI): boot the kernel + billing with **no country pack** loaded and it must fail loudly — no tax rate can exist without a jurisdiction. If it ever "worked", Spain would have leaked into the core. It fails, exactly as required.

What we have — built & verified

COMPONENTS SHIPPED

Kernel

Money (integer, no floats), effective-dating, port registry, spec resolver, events, stores

Quoting + Billing

Immutable invoices, gapless numbering, rectificativas, tax via ports — zero country knowledge

es-ES pack

Effective-dated IVA, renovation reduced-rate engine, IRPF, SHA-256 invoice chain

reformas pack

Mediciones math, partida vocabulary, the construction→tax data contract

Factory + CLI

resolve · validate · new-tenant · demo — composes a tenant from its spec

Database + Web

Postgres + per-tenant row-level security; a first web workspace per tenant

THE WORKING SLICE — lead → presupuesto → factura → get paid



COMPLIANCE ENCODED (RESEARCHED, CITED)

- IVA 10% renovation** — 3 cumulative conditions (recipient, ≥2yr age, ≤40% materials); missing data ⇒ safe higher rate.
- Verifactu** — deadline verified: **2027**. Chaining built; certified submission hard-gated off.
- IRPF retention** — profile-based (most reformas subcontractors ⇒ none).

VERIFIED IN CI ON EVERY PUSH

- Lint · types · unit tests · build · **architecture boundaries**.
- Postgres + RLS** contract tests — cross-tenant reads proven invisible.
- End-to-end browser test of the tenant workspace.

Honest caveat. Tax logic is **defensible by design** (every decision carries its legal basis) but marked `legally_verified: false` until a Spanish *asesor* signs off. No real invoice, email, filing, or payment is ever sent — integrations run as safe fakes behind ports until credentials and sign-off exist. See Page 5.

What's planned — the road to 1,000 tenants

P0

Foundation

Layers, boundary
linter, spec
composition, ADRs

P1

Walking skeleton

CLI, tenant #1,
presupuesto→factura,
negative test

P2

Packs & UI

Persistence ✓, web
shell ✓ · filings,
certificaciones, forms
next

P3

Pipeline

Provision → deploy →
verify → rollback;
PWA / desktop shells

P4

Fleet

Control plane, wave
rollout, health gates,
backup drill

P5

Proof

Timed tenant #2/#3,
offboarding export,
hardening

IMMEDIATE NEXT (P2 CONTINUED)

- Presupuesto forms + line editor with mediciones; PDF export of the factura.
- es-ES filings data: tag data so Modelo 303/390/347/111/115/190 derive automatically; N43 bank-statement import; working-hours register.
- Certificaciones: quoted-vs-actual per partida — where a reforma's margin is won or lost.

DEFINITION OF DONE — THE 5 STOP CONDITIONS

- make demo on a clean machine → legally-defensible factura
- Tenant #2 in <15 min, config-only (mechanism proven at 0.02s)
- Negative test: no country pack ⇒ loud failure
- P0-P5 complete, CI green, no half-migrations
- Governance docs honest + path to tenant #1000 & country #2

The metric that decides success is the **marginal cost of tenant #N+1, in human minutes** — published and tracked from tenant #1. A flat or rising line is treated as a serious defect. Today it is near-zero for a same-country, same-trade tenant; the formal, timed proof (with a real intake) lands in P5.

What input is needed — and when

The build runs unattended on safe defaults — synthetic company data, conservative tax choices, fake integrations. Each item below **replaces a default** with reality. The **When** column is the point at which it stops being optional. Red rows are **gates before any real production use**.

INPUT NEEDED FROM YOU	WHY IT MATTERS	WHEN NEEDED	SAFE DEFAULT UNTIL THEN
Real tenant #1 profile company, CIF/NIF, address, invoice-series habits, languages	Overwrites the synthetic company on every invoice & document	Before go-live	Synthetic "Reformas Iberia Demo S.L.", every field marked <code>source: synthetic</code> — overwrites config only, no code change
Asesor / gestor sign-off on tax logic IVA renovation, IRPF, invoice content	Flips <code>legally_verified: false</code> → <code>true</code> ; required before any real invoice	▲ Gate · go-live	Conservative by design (missing data ⇒ higher rate); decisions stored with legal basis for review
Verifactu decision + AEAT credentials	Enables certified, submitted billing (anti-fraud law)	▲ Before 2027	Invoice chaining + immutability already built; certified submission hard-gated off
Gestor's accounting format A3 / Holded / Sage / CSV	Shapes the accounting-export adapter the gestor consumes	P2 (filings)	Generic CSV export; filing data already tagged on each invoice
Hosting & EU region + provider	Data residency / RGPD; where the fleet runs	P3 (deploy)	Nothing provisioned — the mandate forbids buying paid resources autonomously
Bank access for N43 (or manual upload)	Cash-in reconciliation — cheapest visibility win	P2 (filings)	Fixture-based import behind the bank port
Structural <i>rehabilitación</i> projects?	Unlocks a distinct reduced-rate path (art. 20.Uno.22º.B)	If relevant	Conservative fallback to the general rate with an explanation
Real supplier profiles autónomo empresarial vs. professional	Correct IRPF retention on subcontractor invoices	Go-live	Default "business" ⇒ no retention (the common reformas case)
Public-sector clients? (Facturae / FAcE)	Legally required e-invoice format for the ~5% public mix	If serving public	Not built yet — flagged in the integrations register

☐ Needed soon — unblocks the next phase ☒ ▲ Legal gate — required before real production use ☐ Only if relevant to the business

The one thing to send first: the **real company profile for tenant #1** (name, CIF, address, invoice-series preferences). Everything else has a working default and can follow. Legal sign-off and hosting are the two **gates** that must clear before a first real customer — neither blocks continued building.